

REJOINDER: EM-BASED LIKELIHOOD INFERENCE FOR SOME LIFETIME DISTRIBUTIONS BASED ON LEFT TRUNCATED AND RIGHT CENSORED DATA AND ASSOCIATED MODEL DISCRIMINATION

N. Balakrishnan¹

McMaster University, Canada

e-mail: *bala@mcmaster.ca*

and

Debanjan Mitra

Indian Institute of Management Udaipur, India

First of all, we express our sincere thanks to Drs. Laurent Bordes of Université de Pau et des Pays de l'Adour and Didier Chauveau of Université d'Orléans, Dr. Isha Dewan of Indian Statistical Institute at New Delhi, Drs. Hon Keung Tony Ng of Southern Methodist University and Zhisheng Ye of Hong Kong Polytechnic University, Drs. Yili Hong and Caleb King of Virginia Tech, Drs. Iain L. MacDonald and Brendon M. Lapham of University of Cape Town, Dr. Tertius de Wet of Stellenbosch University, and Dr. Hideki Nagatsuka of Chuo University for writing insightful discussions on our invited paper. Their valuable discussions certainly further the topic of discussion of our paper by providing some additional insight into the topic and also by adding some more directions of future research in the analysis of left truncated and right censored data.

Drs. Ng and Ye and Drs. Bordes and Chauveau have discussed the stochastic-EM algorithm in the context considered in our paper. While Drs. Bordes and Chauveau have discussed the stochastic-EM algorithm for the case of Weibull lifetime distribution, Drs. Ng and Ye have developed the stochastic-EM algorithm for the generalized gamma distribution, both under left truncated and right censored data. In both these discussions, the stochastic-EM algorithm has been explained clearly, and the specific steps for Weibull and generalized gamma distributions have been developed in a careful and comprehensive manner. For the Weibull distribution, it is seen that the results obtained by Drs. Bordes and Chauveau are quite close to those obtained by us. However, for the generalized gamma distribution with left truncation and right censoring, Drs. Ng and Ye have pointed out that the EM algorithm may converge to a local maxima. In this case, the stochastic-EM algorithm clearly provides a better alternative, as it avoids getting trapped into any saddle point. Our special thanks go to Drs. Ng and Ye for pointing out this issue with the EM algorithm for left truncated and right censored data from the generalized gamma distribution.

Now, with regard to the proposed EM algorithm for left truncated and right censored data from the generalized gamma distribution, it may converge to a local maxima. Indeed, the proposed EM

¹Corresponding author.

algorithm in this particular case is actually the EM gradient algorithm (Lange, 1995), and it is a well known fact (which is also mentioned in the original paper by Lange (1995)) that the EM gradient algorithm depends heavily on the choice of initial values, due to the very nature of the algorithm. In our work, the initial values for the proposed EM algorithm for the generalized gamma case have been chosen through a grid search technique. We could possibly think of some ways to avoid the convergence towards a local maxima. While one possible way would be to check the convergence using several different initial values, a more systematic alternative would be to use a grid search over a greater span. However, since the stochastic EM algorithm, as discussed by Drs. Ng and Ye, clearly seems to avoid this problem, one may use it for the case of the generalized gamma distribution with left truncated and right censored data.

Drs. MacDonald and Lapham have asked three pertinent questions. First of all, they have pointed out that the EM algorithm needs more effort to be applied than the general purpose optimizers readily available in software packages such as `maxNR`. It is, of course, undeniable that the EM algorithm needs much more careful effort – first in deriving the necessary expressions, and then in coding those expressions properly. It should also be pointed out that the numerical methods, such as the Newton-Raphson, have their own disadvantages. They are not guaranteed to converge to the global maxima in the first place and also can have an oscillating behavior during successive iterations. On the other hand, the EM algorithm is based on sound statistical reasoning and understanding of what is going on underneath the estimation procedure, though it surely needs much more effort for its implementation; interested readers may refer to the book by McLachlan and Krishnan (2008) for a comprehensive discussion on its merits and benefits. Furthermore, another key advantage of the EM algorithm is that, with its use in conjunction with the missing information principle, it will enable one to assess the amount of information lost due to censoring in the estimation of the model parameters.

The second question of Drs. MacDonald and Lapham is about the use of the EM gradient algorithm, i.e., the one-step Newton-Raphson, in the M-step. Indeed, in the M-step of the EM gradient algorithm, just one step of the Newton-Raphson algorithm is performed instead of the entire Newton-Raphson process. Thus, the Q -function is just increased, and then the E-step is repeated once again. As the generalized EM algorithm states, if the Q -function can be increased in the subsequent steps, it is a sufficient condition for the convergence; see McLachlan and Krishnan (2008). This is what we have implemented too, and indeed the word “increased” should be used here. We thank them for pointing this out.

The third question posed by them is about the asymptotic confidence intervals considered for the parameters of all the models. We have not considered profile likelihood based confidence intervals for the parameters in our work. As pointed out by them, the profile likelihood based confidence intervals may provide some advantages, and this could be a fruitful area for future research. However, we would like to end our comment on this by mentioning that the profile likelihood approach for estimating the parameters may sometimes bring in some additional complications in the case of analysis of left truncated and right censored data. For example, we did try using the profile likelihood approach for the parameter estimation of the generalized gamma distribution with left truncated and right censored data, but the flatness of the likelihood surface (with respect to the shape parameters) caused some problems. Furthermore, the method also resulted in quite conservative standard errors which in turn resulted in under-coverage of the confidence intervals. Moreover, there are some clear

advantages in using the missing information principle in these situations, as discussed in the paper. For example, through the missing information principle, one can not only obtain the asymptotic confidence intervals, but can also estimate the loss of efficiency in the parameter estimates due to censoring, as mentioned earlier. This, in our opinion, gives some additional valuable insight on the inferential procedure as a whole.

In his discussion, Dr. Tertius de Wet has talked about the estimation of the underlying distribution involved. He presented a careful literature survey on nonparametric maximum likelihood estimators of the distribution function of the lifetime variable and the corresponding quantile function. He has aptly pointed out that a potential direction of future research could be the investigation of goodness-of-fit tests and their properties under the left truncated and right censored data. He has also suggested the investigation of the effect of the estimated parameters of the underlying model on the limiting distributions. In this regard, he has presented an illustrative example that shows an estimate of the distribution function and the form of the resulting Q-Q plots under a correct as well as an incorrect distributional assumption. His comprehensive discussion has certainly enriched our article by furnishing relevant literature surveys and also by suggesting potential future areas of research.

Dr. Isha Dewan has presented a brief discussion on the nonparametric maximum likelihood estimator of the distribution function. She has also mentioned that she and her collaborators have successfully applied the EM algorithm for parameter estimation of bivariate survival data based on the Marshall-Olkin bivariate Weibull distribution and the generalized bivariate exponential distribution, and that they have observed through their study that the EM algorithm performs well even when censoring is present in the data. She has provided some useful references in various directions related to left truncated and right censored data concerning the nonparametric maximum likelihood estimation of the distribution function, nonparametric estimation of the bivariate survival function when both components of the random vector are subject to left truncation and right censoring, and the analysis of competing risks data.

A very careful and diligent discussion of our work has been made by Drs. Hong and King. It is important to mention here that our work on the likelihood inference for left truncated and right censored data got initiated by the pioneering work of Hong, Meeker and McCalley (2009). Our sincere thanks go to them for their kind attention to our work.

It is interesting to see the alternative way of modelling that Dr. Nagatsuka has suggested in his discussion. He has suggested a penalized EM algorithm and, through a detailed discussion, has demonstrated how the method can reduce the instability in the likelihood estimation procedure for the case of the Weibull distribution when the shape parameter is large. It will certainly be of interest for a future research project to investigate how the proposed penalized EM algorithm works for other lifetime distributions as well based on left truncated and right censored data.

In concluding, we thank all the discussants for their incisive comments and useful suggestions on some related problems and possible further work. This has most certainly complemented our discussion on this topic of research and have supplied much more food for thought. Our final thanks go to Professor Maxim Finkelstein, the Editor-in-Chief, for inviting us to prepare this review article and then furnishing a platform for having it discussed by a group of such active and distinguished researchers. It is our sincere hope that this discussion paper will help generate a lot more additional valuable research work on this topic!

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